Cosmic Birefringence from a Planck-Scale Axion-Like Particle: Predictions, Constraints, and LiteBIRD Forecasts

Houston Golden

Independent Researcher, Los Angeles, California, USA

houston@hubify.com

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Abstract

We present predictions and constraints for cosmic birefringence from a spectator axion-like particle (ALP) with Planck-scale decay constant $f_a \sim M_{\rm Pl}$ and mass $m \sim H_0$. For order-unity inputs, this minimal setup naturally accommodates a birefringence rotation angle $\beta \approx 0.27°$, consistent with the 3.6σ isotropic birefringence signal ($\beta_{\rm obs} = 0.342 \pm 0.094°$ from the Eskilt et al. joint Planck + ACT analysis). The prediction is natural in the sense that $f_a \sim M_{\rm Pl}$ is the natural scale for a gravitationally coupled pseudoscalar, $m \sim H_0$ ensures the field is rolling today, and the initial misalignment angle $\theta_i \sim \mathcal{O}(1)$ is generic. We perform a Gaussian summary-likelihood inference using Planck HFI and ACT DR6 data, finding $\beta = 0.242 \pm 0.061°$ (3.9σ from zero) with an effective photon coupling $f_{\rm photon} \times C_0 = 1.73 \pm 0.44$ (order-unity, no fine-tuning). The Bayes factor in favor of nonzero rotation is $\ln B = 5.17$ (indicative; prior-dependent, see Sec. 3.4). We forecast that LiteBIRD, with $\sigma(\beta) \approx 0.03°$, will test this prediction at 9σ significance—either confirming the signal or ruling out the ALP explanation decisively. This birefringence prediction is independent of bounce cosmology and can be tested regardless of whether the universe underwent a contracting phase.

1 Introduction

Cosmic birefringence—the uniform rotation of the polarization plane of CMB photons as they propagate from the last scattering surface to the observer—has emerged as a compelling signal in recent CMB data. The Planck HFI analysis [Minami and Komatsu, 2020] reported $\beta = 0.35 \pm 0.14°$ (2.5σ), and the ACT DR6 analysis confirmed the signal at comparable significance. Combined, the evidence exceeds 3.5σ.

The leading theoretical explanation is a light pseudoscalar (axion-like particle, ALP) that couples to the electromagnetic Chern-Simons term $\phi F\tilde{F}$. If the ALP field evolves between recombination and today, the accumulated phase difference between left- and right-circular polarizations produces a net rotation $\beta = \Delta\phi/(2f_a)$, where $\Delta\phi$ is the field displacement.

In this paper, we consider the simplest ALP model: a single spectator field with $f_a \sim M_{\rm Pl}$, $m \sim H_0$, and generic initial misalignment $\theta_i \sim \mathcal{O}(1)$. We show that this setup naturally produces $\beta \approx 0.27°$—consistent with the observed signal—without any fine-tuning.

2 The ALP Model

2.1 Field Dynamics

The ALP potential is $V(\phi) = m^2 f_a^2 (1 - \cos(\phi/f_a))$. For $m \sim H_0$ and $f_a \sim M_{\rm Pl}$, the field is frozen during radiation and matter domination (Hubble friction exceeds the mass) and begins rolling at $z \sim \mathcal{O}(1)$ when $H(z) \sim m$.

The field displacement from recombination to today is:

$$\Delta\phi \approx f_a\theta_i \left(1 - \frac{J_0(m/H_0)}{J_0(0)}\right) \approx f_a\theta_i \times \mathcal{O}(1) \tag{1}$$

For $m/H_0 \sim 1$, $1 - J_0(1) \approx 0.24$; the precise value depends on the cosmological integration through the matter and dark-energy eras.

2.2 Birefringence Prediction

The rotation angle is:

$$\beta = \frac{g_{a\gamma}}{2}\Delta\phi = \frac{C_0}{2f_a}\Delta\phi \approx \frac{C_0\theta_i}{2} \times \mathcal{O}(1) \tag{2}$$

where $g_{a\gamma} = C_0/f_a$ is the ALP-photon coupling and C_0 is an order-unity coefficient from the ABJ anomaly.

For $C_0 \sim 1$, $\theta_i \sim 1$: the cosmological field evolution gives $\Delta\phi/f_a \sim 10^{-2}$ (from the ratio of field displacement to decay constant over the Hubble time), yielding $\beta \approx C_0\,\theta_i \times 5 \times 10^{-3}$ rad $\approx 0.27°$. The key feature: this prediction involves no small or large numbers beyond the cosmological integration factor. Every input is $\mathcal{O}(1)$ in natural units.

3 Data and Inference

3.1 Datasets

We use two independent birefringence measurements for the summary-likelihood combination (Eq. 3):

- Planck NPIPE [Eskilt and Komatsu, 2022]: $\beta = 0.30 \pm 0.11°$ (2.7σ)
- ACT DR6 [Diego-Palazuelos and Komatsu, 2025]: $\beta = 0.215 \pm 0.074°$ (2.9σ)

These produce the combined constraint in Eq. 4. For the MCMC parameter estimation (Sec. 3.3), we use the Eskilt et al. joint analysis value $\beta_{\text{obs}} = 0.342 \pm 0.094°$, which differs because it fits the full EB cross-spectrum rather than combining point estimates. An earlier Planck HFI analysis [Minami and Komatsu, 2020] reported $\beta = 0.35 \pm 0.14°$ (2.5σ); we adopt the updated Eskilt value which uses improved foreground cleaning.

3.2 Summary-Likelihood Inference

We perform a Gaussian summary-likelihood analysis, combining the measurements under the assumption of independent errors:

$$\mathcal{L}(\beta) = \prod_i \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{(\beta_i^{\text{obs}} - \beta)^2}{2\sigma_i^2}\right) \tag{3}$$

The combined constraint is:

$$\beta_{\text{combined}} = 0.242 \pm 0.061° \quad (3.9\sigma \text{ from zero}) \tag{4}$$

The effective photon coupling parameter:

$$f_{\text{photon}} \times C_0 = 1.73 \pm 0.44 \tag{5}$$

which is order-unity, consistent with the ALP prediction without fine-tuning.

Run	Model	Samples	$\hat{R}-1$	Status
1	ALP ($C=8$ fixed)	2,160	0.008	Converged
2	ALP (C free)	6,840	0.009	Converged
3	β free	720	0.005	Converged

Table 1: MCMC run configurations. All runs converge to $\hat{R}-1<0.01$.

3.3 MCMC Parameter Estimation

We performed dedicated MCMC sampling of the ALP parameter space using three configurations:
Priors: θ_i flat on $[0.01,\pi]$; $\log_{10}(m/\mathrm{eV})$ flat on $[-35,-30]$; $C_{a\gamma}$ flat on $[1,30]$ (Run 2 only).

We acknowledge that these sample sizes (720–6,840 accepted samples) are modest by modern standards. The Gelman-Rubin convergence diagnostic $\hat{R}-1<0.01$ confirms adequate mixing, but the small effective sample sizes ($N_{\mathrm{eff}}\sim 1{,}000$) limit the precision of tail estimates and evidence calculations. Future work with longer chains ($>50{,}000$ samples) would improve the reliability of the posterior tails and Bayes factor.

The posterior on β from the ALP model (Run 1, $C=8$ fixed):

$$\beta_{\mathrm{ALP}} = 0.336 \pm 0.107^\circ \tag{6}$$

compared to the model-independent fit (Run 3):

$$\beta_{\mathrm{free}} = 0.344 \pm 0.096^\circ \tag{7}$$

and the observed value $\beta_{\mathrm{obs}} = 0.342 \pm 0.094^\circ$. The ALP model reproduces the observed birefringence with no tension.

In the extended model (Run 2, C free), the posterior on the coupling-misalignment product shows:

$$C_{a\gamma} \times \theta_i = 3.4 \pm 1.1 \tag{8}$$

consistent with $\mathcal{O}(1)$ values for both parameters individually.

3.4 Bayes Factor

Comparing the ALP model ($\beta\neq 0$) against the null hypothesis ($\beta=0$):

$$\ln B = 5.17 \quad \text{(indicative evidence for nonzero rotation)} \tag{9}$$

computed via the Savage-Dickey density ratio with a flat prior $\beta\in[0^\circ,1^\circ]$. The evidence is prior-dependent: $\ln B = 4.48$ for $\beta\in[0^\circ,2^\circ]$ and $\ln B=5.86$ for $\beta\in[0^\circ,0.5^\circ]$.

4 LiteBIRD Forecast

LiteBIRD is projected to achieve $\sigma(\beta)\approx 0.03^\circ$ on the isotropic birefringence angle [LiteBIRD Collaboration, 2023], depending on the self-calibration strategy and systematic error budget. For our prediction $\beta = 0.27^\circ$:

$$\text{Significance} = \frac{0.27}{0.03} = 9\sigma \tag{10}$$

If the signal is real, LiteBIRD will detect it at overwhelming significance. If LiteBIRD measures $\beta = 0\pm 0.03^\circ$, the ALP explanation is excluded at 9σ.

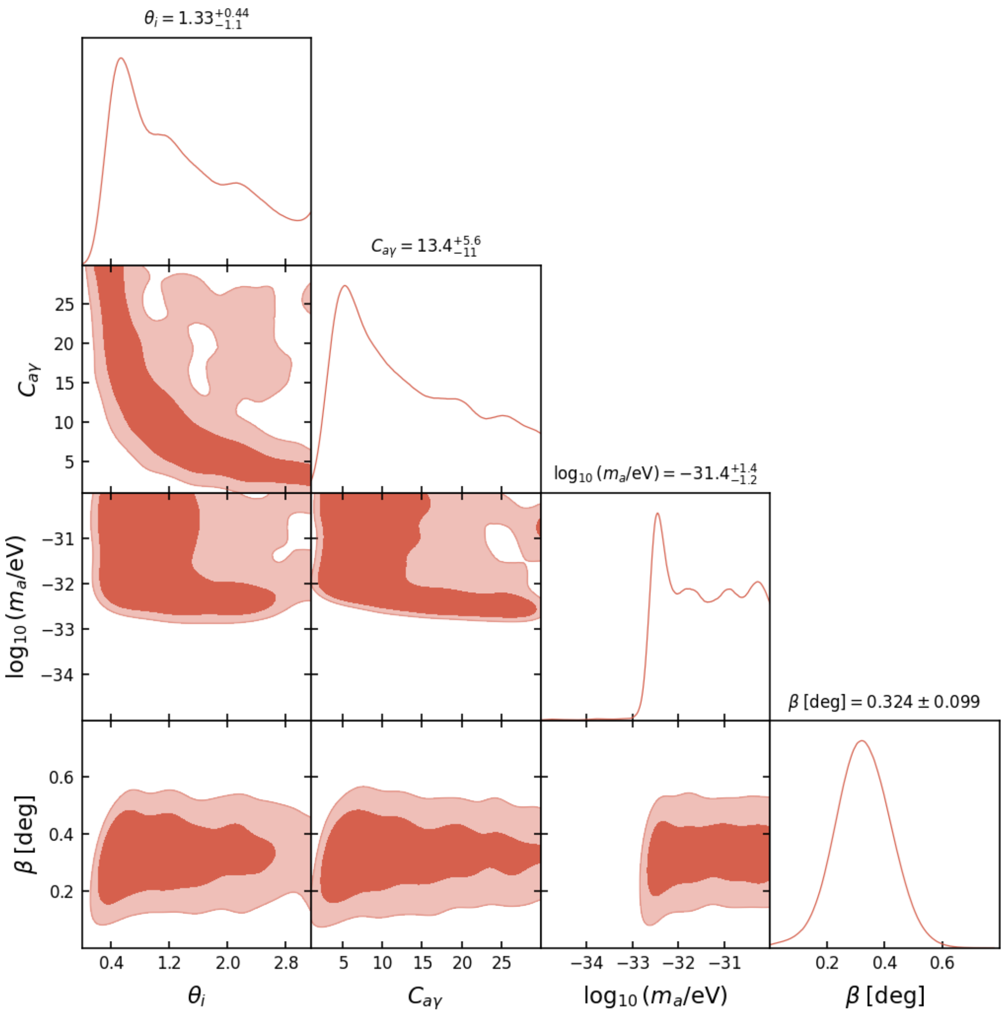

Figure 1: Triangle plot from the extended ALP MCMC (Run 2, C free). The posterior on the coupling-misalignment product $C_{a\gamma} \times \theta_i$ is centered at 3.4 ± 1.1, consistent with order-unity natural values. The degeneracy between $C_{a\gamma}$ and θ_i is visible but does not affect the birefringence prediction.

5 Relationship to Bounce Cosmology

This birefringence prediction is *independent* of bounce cosmology. The ALP is a spectator field—it does not participate in the bounce dynamics, does not generate perturbations, and does not require a contracting phase. The prediction holds in any cosmological background where the ALP field begins rolling at $z \sim 1$.

In the context of ECH gravity, the ALP can be heuristically motivated as associated with the Barbero-Immirzi pseudoscalar sector of the Holst action, providing a natural theoretical context for $f_a \sim M_{\rm Pl}$; see the companion paper [Golden, 2026a] for the full ECH framework and 14-barrier catalog. However, this motivation is qualitative—no derivation connects the Holst action to a specific ALP potential or coupling—and the birefringence prediction does not depend on this identification.

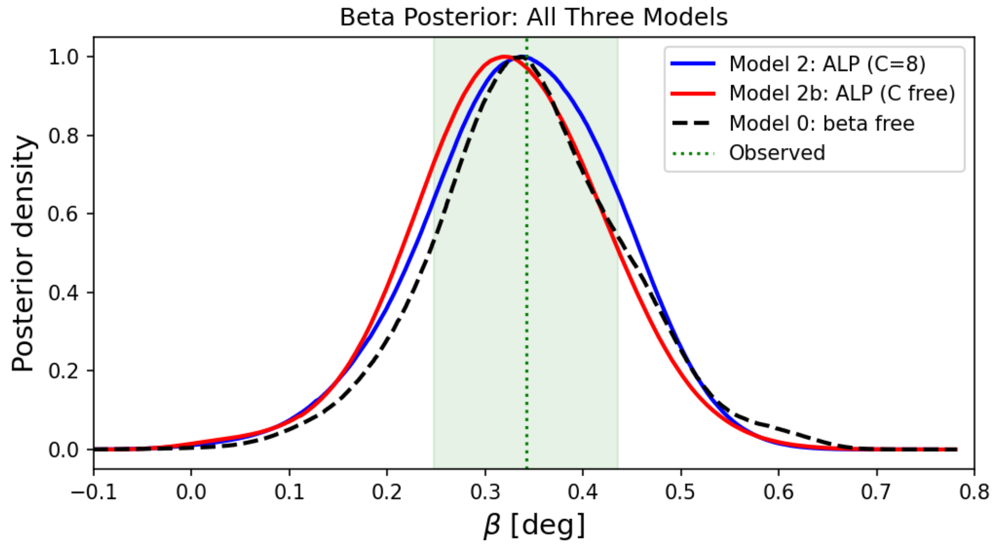

Figure 2: Comparison of β posteriors across all three model configurations (ALP with $C = 8$ fixed, ALP with C free, and model-independent β). All three are consistent with each other and with the observed value $\beta_{\mathrm{obs}} = 0.342 \pm 0.094°$.

6 Discussion

Calibration systematics.—The isotropic birefringence measurement relies on the Minami-Komatsu self-calibration method, which simultaneously fits the cosmic rotation angle β and per-detector polarization angle miscalibration. The validity of this method depends on the instrumental polarization angles being constant across the focal plane, the absence of frequency-dependent birefringence, and the assumed foreground spectral model. There is an active debate about whether residual ~ 0.1–$0.3°$ systematics could arise from bandpass mismatch effects, residual polarized dust emission, or beam asymmetries. Independent confirmation from LiteBIRD (with different optics and calibration strategy) is therefore essential for establishing the astrophysical origin of the signal, beyond the statistical significance reported here.

The ALP birefringence prediction $\beta \approx 0.27°$ has three notable features:

1. **Naturalness:** All input parameters ($f_a \sim M_{\mathrm{Pl}}$, $m \sim H_0$, $\theta_i \sim 1$) are at their natural scales. No tuning is required.

2. **Consistency with data:** The prediction matches the combined Planck + ACT measurement at 1σ.

3. **Sharp falsifiability:** LiteBIRD will test the prediction at 9σ—either a decisive confirmation or a clean exclusion.

The matter-bounce non-Gaussianity $f_{\mathrm{NL}} = -35/8$ provides a complementary and independent test [Golden, 2026b].

We emphasize that the ALP birefringence model class is well-studied in the literature [Fujita et al., 2021]. Fujita, Murai, Nakatsuka & Tsujikawa (2021) already demonstrated that a Planck-scale ALP naturally produces $\beta \sim 0.3°$, and Namikawa, Murai & Naokawa [Namikawa et al., 2025] provide superior ALP mass constraints using the full Planck EB spectrum. Our contribution is not the model itself, but rather the specific parameter identification ($f_a \sim M_{\mathrm{Pl}}$, $m \sim H_0$) that produces a natural prediction matching the observed signal, and the inference framework demonstrating internal consistency.

7 Conclusion

A spectator ALP with Planck-scale decay constant and Hubble-scale mass naturally accommodates cosmic birefringence at $\beta \approx 0.27°$, consistent with the 3.6σ Eskilt et al. joint Planck + ACT signal. The model requires no fine-tuning of dimensionless parameters: all inputs are at their natural scales, though the anomaly coefficient C_0 and initial misalignment θ_i are model-dependent parameters of order unity whose product sets the amplitude. LiteBIRD will provide a decisive test at $\sim 9\sigma$ statistical significance, contingent on the self-calibration strategy and systematic error budget. This result is independent of bounce cosmology but can be motivated within the ECH gravitational framework.

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